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A High-Efficiency and Robust Compressive Sensing Reconstruction and Regularization Technique

Lieqian Dong, BGP, CNPC, Zhuozhou, China

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Abstract

Seismic data acquisition is a crucial procedure in seismic exploration. Compressive sensing (CS) is a new signal sampling framework and breaks through the limitation of Nyquist theorem (Donoho, 2006). The application of CS in seismic data acquisition (Mosher et al. 2012) can potentially improve exploration efficiency and lower costs. CS helps the reconstruction of seismic data before subsequent data processing stages. Numerous researchers have made contributions to the advancement of techniques for reconstructing regularly sampled data. The majority of these methods fall under the category of prediction error filter techniques (Li et al. 2017; Spitz 1991; Zheng et al. 2022), sparse transformation approaches (Fomel & Liu 2010; Gong et al. 2016), and low-rank constraint approaches (Sacchi 2009). In recent years, numerous deep learning-based reconstruction methods have been proposed (Chen et al. 2023; Liu & Ma 2024). These methods are often used to uniformly sampled data. In non-uniform sampling cases, the data are usually binning to a uniform grid. However, data binning ignores the actual positions of seismic traces, and compromises the continuity of seismic events.

To address the challenges presented by the distortions and artifacts caused by the binning methods, researchers have explored non-uniform data processing methods using specialized non-uniform sampling transforms. Notable among these are non-uniform Fourier transform (Yu et al. 2020), and non-uniform curvelet transform (Zhang et al. 2019). Kumar et al. (2015) extended the matrix completeness to a non-uniform grid to conduct regularization of two-dimensional seismic data. Jiang et al. (2017) extended the traditional projection onto convex sets (POCS) approach (Abma & Kabir 2006) to the extended POCS (EPOCS), which incorporates an interpolation operator and broadens its applicability to arbitrary irregular acquisition. Carozzi & Sacchi (2021) modified the multi-singular spectrum analysis (MSSA) and proposed an interpolated MSSA (I-MSSA) algorithm, which is a reconstruction algorithm that honors real coordinates. Li et al. (2012) proposed a 2D seismic data reconstruction approach relying on compressed sensing interpolation. Yu et al. (2020) utilized non-uniform Fourier transform to map seismic data from uniform to non-uniform grids, and applied this approach for 3D seismic data reconstruction. Yu & Ma (2023) proposed a new combined sampling operator that can simultaneously perform non-uniform grid regularization and reconstruction of 3D seismic data. This method reconstructs 3D seismic data with a 2D barycentric Lagrangian (2DBL) interpolator (Berrut & Trefethen 2004), which significantly reduces the

computational cost. In field seismic exploration, the sampled data may contain random and outlier noise. Outlier noise refers to values or signals in a dataset that significantly deviate from other data points (Osborne & Overbay 2019). These outliers may arise due to measurement errors, sensor malfunctions, environmental interference, or other abnormal conditions. Random noise refers to small, unpredictable changes scattered throughout the data, often due to minor factors like measurement errors. Numerous methods have been introduced to suppress the random noise in seismic data (Zeng et al. 2022). In seismic data processing, the identification and handling of outlier noise are crucial, as such noise can distort the characteristics of the actual seismic data. Zhao et al. (2018) proposed a robust curvelet thresholding approach by formulating an optimization problem using Huber norm regularization. Li et al. (2019) developed a POCS method enhanced with Huber- optimization for seismic data interpolation. Chen et al. (2022) developed a robust deep learning (DL) denoising approach for seismic data that operates in a self-supervised way without clean data.

In the paper, a high-efficiency and robust seismic data reconstruction method based on fast sparse inversion is proposed. Compared with ISTA, the main highlight of the new method is the adoption of an accelerated iterative shrinkage threshold algorithm (AISTA) to improve the convergence speed and reconstruction accuracy. Considering the existence of the obstacle in field operation, a non-uniform interpolator operator is designed to alleviate the adverse influence to reconstruction accuracy. Moreover, 3D curvelet transform (Candes et al., 2004; Herrmann et al., 2008) is selected as sparse operator to boost the iterative convergence speed. In order to decrease the computational redundancy of 3D curvelet transform, a parallel patching strategy is utilized to segment a large volume of data into small patches, following to parallel-compute the 3D curvelet coefficient of each small patch to further improve the computational efficiency. Finally, we demonstrate the effectiveness of the proposed method by massive field datasets.

Introduction

Seismic data acquisition is a crucial procedure in seismic exploration. Compressive sensing (CS) is a new signal sampling framework and breaks through the limitation of Nyquist theorem (Donoho, 2006). The application of CS in seismic data acquisition (Mosher et al. 2012) can potentially improve exploration efficiency and lower costs. CS helps the reconstruction of seismic data before subsequent data processing stages. Numerous researchers have made contributions to the advancement of techniques for reconstructing regularly sampled data. The majority of these methods fall under the category of prediction error filter techniques (Li et al. 2017; Spitz 1991; Zheng et al. 2022), sparse transformation approaches (Fomel & Liu 2010; Gong et al. 2016), and low-rank constraint approaches (Sacchi 2009). In recent years, numerous deep learning-based reconstruction methods have been proposed (Chen et al. 2023; Liu & Ma 2024). These methods are often used to uniformly sampled data. In non-uniform sampling cases, the data are usually binning to a uniform grid. However, data binning ignores the actual positions of seismic traces, and compromises the continuity of seismic events.

To address the challenges presented by the distortions and artifacts caused by the binning methods, researchers have explored non-uniform data processing methods using specialized non-uniform sampling transforms. Notable among these are non-uniform Fourier transform (Yu et al. 2020), and non-uniform curvelet transform (Zhang et al. 2019). Kumar et al. (2015) extended the matrix completeness to a non-uniform grid to conduct regularization of two-dimensional seismic data. Jiang et al. (2017) extended the traditional projection onto convex sets (POCS) approach (Abma & Kabir 2006) to the extended POCS (EPOCS), which incorporates an interpolation operator and broadens its applicability to arbitrary irregular acquisition. Carozzi & Sacchi (2021) modified the multi-singular spectrum analysis (MSSA) and proposed an interpolated MSSA (I-MSSA) algorithm, which is a reconstruction algorithm that honors real coordinates. Li et al. (2012) proposed a 2D seismic data reconstruction approach relying on compressed sensing interpolation. Yu et al. (2020) utilized non-uniform Fourier transform to map seismic data from uniform to non-uniform grids, and applied this approach for 3D seismic data reconstruction. Yu & Ma

(2023) proposed a new combined sampling operator that can simultaneously perform non-uniform grid regularization and reconstruction of 3D seismic data. This method reconstructs 3D seismic data with a 2D barycentric Lagrangian (2DBL) interpolator (Berrut & Trefethen 2004), which significantly reduces the computational cost. In field seismic exploration, the sampled data may contain random and outlier noise. Outlier noise refers to values or signals in a dataset that significantly deviate from other data points (Osborne & Overbay 2019). These outliers may arise due to measurement errors, sensor malfunctions, environmental interference, or other abnormal conditions. Random noise refers to small, unpredictable changes scattered throughout the data, often due to minor factors like measurement errors. Numerous methods have been introduced to suppress the random noise in seismic data (Zeng et al. 2022). In seismic data processing, the identification and handling of outlier noise are crucial, as such noise can distort the characteristics of the actual seismic data. Zhao et al. (2018) proposed a robust curvelet thresholding approach by formulating an optimization problem using Huber norm regularization. Li et al. (2019) developed a POCS method enhanced with Huber- optimization for seismic data interpolation. Chen et al. (2022) developed a robust deep learning (DL) denoising approach for seismic data that operates in a self-supervised way without clean data.

In the paper, a high-efficiency and robust seismic data reconstruction method based on fast sparse inversion is proposed. Compared with ISTA, the main highlight of the new method is the adoption of an accelerated iterative shrinkage threshold algorithm (AISTA) to improve the convergence speed and reconstruction accuracy. Considering the existence of the obstacle in field operation, a non-uniform interpolator operator is designed to alleviate the adverse influence to reconstruction accuracy. Moreover, 3D curvelet transform (Candes et al., 2004; Herrmann et al., 2008) is selected as sparse operator to boost the iterative convergence speed. In order to decrease the computational redundancy of 3D curvelet transform, a parallel patching strategy is utilized to segment a large volume of data into small patches, following to parallel-compute the 3D curvelet coefficient of each small patch to further improve the computational efficiency. Finally, we demonstrate the effectiveness of the proposed method by massive field datasets.

Theory

The compressive sensing seismic data acquisition can be mathematically expressed in the following form,

$$\boldsymbol{Y}^{obs} = \boldsymbol{P}\boldsymbol{Y} + \boldsymbol{n} \tag{1}$$

where $\boldsymbol{Y}_{obs}$ is the observered data on irregular grid, $\boldsymbol{Y}$ denotes the expected complete data on regular grid, $\boldsymbol{n}$ is the noise, $\boldsymbol{P}$ is the sampling operator which contains 0 for missing data and 1 for observered data. The purpose of the seismic data reconstruction is to recover $\boldsymbol{Y}$ on regular grid from $\boldsymbol{Y}_{obs}$ on irregular grid.

To solve the ill-posed nature of the underdetermined problem in equation 1, an extra constraint is used to promote the divergence of the solution. Therefore, we build the following objective function:

$$\min_{x} \| \boldsymbol{A}\boldsymbol{x} - \boldsymbol{Y}^{obs} \|_2^2 + \alpha g(\boldsymbol{x}) \tag{2}$$

where $A = PC^T$, $x = CY$, $\| \|_2^2$ is the L2 norm, $\alpha g(x)$ is the shaping regularization term to boost the convergence of solving the under-determined problem, C is a sparsity-promoting transform. In the paper, 3D curvelet transform is selected.

The under-determined inversion problem is usually solved by an iterative shrinkage threshold algorithm (ISTA) in sparse transform domain. The general expression of ISTA is given as follows,

$$Y_{k+1} = (I - P)C^T T_{\lambda} C Y_k + Y^{obs} \tag{3}$$

where Y_k denotes the reconstruction result after the k_{th} iteration, C^{-1} is the inversion of C, I is unit matrix, T_{λ} represents the threshold function with threshold value decreases for each iteration. In the paper, the hard threshold function is selected.

The conventional ISTA based sparse inversion methods face the challenge of low conputaitonal efficiency, and can not satisfy the need of massive CS data processing. Therefore, two optimal strategies are proposed to improve reconstruction efficiency and accuracy.

Strategy 1: Accelerated iterative shrinkage threshold algorithm (AISTA)

In order to improve the iterative efficiency of ISTA, the AISTA is proposed on the basis of the research of Beck et al. in 2009 and adopts two-step optimization scheme to boost the convergence speed and reconstruction accuracy. The AISTA can be expressed as:

$$
\begin{aligned}
Y_{k+1} &= X_k + \frac{t_k - 1}{t_{k+1}}\left(X_k - X_{k-1}\right) \\
Z_{k+1} &= (I\text{-P})Y_{k+1} + PY^{obs} \\
X_k &= C^T T_\lambda C\left(Z_k\right) \\
t_{k+1} &= \frac{1+\sqrt{1+4t_k^2}}{2}.
\end{aligned} \tag{4}
$$

where

Figure 1 shows the change of reconstruction signal-to-noise (S/N) ratio with iterations by adopting AISTA method and ISTA method respectively. In the test, the same simulated data with randomly decimated by 50% is selected. From the comparisons, we can see the AISTA method has faster convergence speed and reconstruction accuracy than using ISTA method. In this way, the reconstruction accuracy and computational efficiency will be improved at the same time which is vital importance in large-scale seismic data processing.

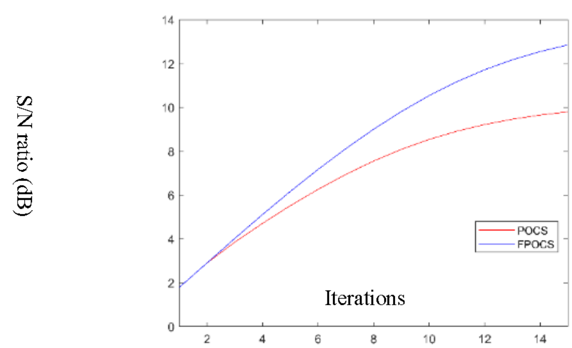

Figure 1—The change of reconstruction S/N ratio with iterations (the blue curve is the AISTA method and the red curve is the ISTA method)

Strategy 2: Parallel patching computing

Comparing with FFT and wavelet transform, 3D curvelet transform offers a highly sparse representation of a wide range of 3D signals and images. Sparsity means that a large number of the curvelet coefficients are close to zero, and only a relatively small number of non – zero coefficients are sufficient to accurately reconstruct the original 3D data. This property is rooted in the ability of curvelets to match the geometric structures present in the data. However, the computational redundancy limits the application of 3D curvelet transform. In the paper, a parallel patching strategy is utilized to segment a large volume of data into small patches, following to parallel-compute the 3D curvelet coefficient of each small patch to decrease the redundancy and improve the efficiency. Figure 2 shows the diagram of the parallel patching strategy. The large volume of data is segmented into some patches along time axis according the computing resource configuration and seismic data size.

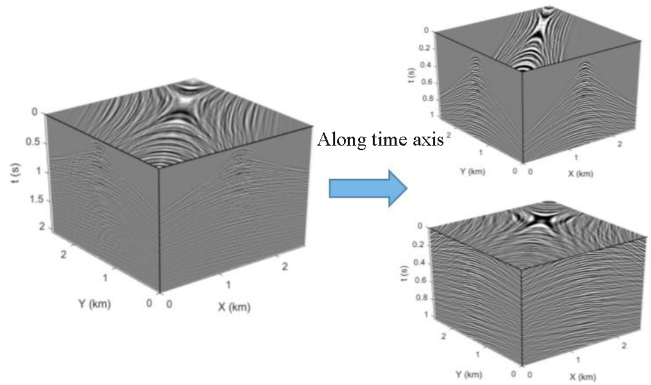

Figure 2—The diagram of the parallel patching to segment a large volume of data into small patches

Strategy 3: non-uniform interpolation operator

Because field construction conditions can also disrupt the sampling geometry, resulting in missing samples and samples off-the-grid to the designed grid, which will affect the reconstruction accuracy, we treat missing and off-the-grid samples separately and propose a new sampling operator that combines the barycentric Lagrangian (BL) operator and a binary mask (BM).

The non-uniform interpolator calculates values at specific points by using a weighted sum of data points that are located on the uniform grid. We take a time-slice data example, the observed data is denoted as $b(x_i,y_i)$. The data associated with the uniform grid points is represented by $z(x_k, y_k)$, where (x_k, y_k) defines the uniform grid. It is assumed that each sample coordinate (x_i, y_i) is encompassed by some uniform grid point (x_k, y_k). The observed data is located on this grid, the form is shown in equation (5).

$$b(x_i, y_i) = \sum_{k\in\mathcal{N}_i} A_{i,k} z(x_k, y_k). \tag{5}$$

Where $\mathcal{N}_i$ represents a uniform set of grid points around (x_i,y_i), $A_{i,k}$ is the 2DBL interpolation weight. In this context, the forward interpolation operator takes multiple uniform samples $z(x_k, y_k)$ and combines them into a single non-uniform sample value using a weighted sum. Conversely, the adjoint interpolation operator disperses a single non-uniform value to multiple uniform points based on the corresponding weights (Carozzi & Sacchi 2021).

$$\tilde{z}(x_k, y_k) = A_{i,k} b(x_i, y_i) \quad k \in \mathcal{N}_i. \tag{6}$$

The 2D BL interpolation operator is defined explicitly as follows

$$A_{i,k} = \frac{\omega_k}{(x_i - x_k)(y_i - y_k)} \Bigg/ \left(\sum_{j\in\mathcal{N}_i} \frac{\omega_j}{(x_i - x_j)(y_i - y_j)}\right), \tag{7}$$

where (x_k, y_k) is the uniform grid coordinate, (x_i, y_i) is the non-uniform grid coordinate. ω_k is the barycentric weight which defined as follows:

$$\omega_k = \frac{1}{\prod_{l=1,l\neq k}^{n}(x_k - x_l)\prod_{m=1,\, m\neq k}^{n}(y_k - y_m)}, \tag{8}$$

Data tests

In this section, we verify the effectiveness of the proposed approach through a simulated and a field CS datasets. The signal-to-noise ratio (SNR) is employed to measure the quality of reconstruction results, with is defined in Equation (9).

$$\mathrm{SNR} = 10\log\frac{\|D\|_F^2}{\|D - D_{rec}\|_F^2}, \tag{9}$$

where D is the complete data and D_{rec} is the reconstructed data.

The first 3D synthetic dataset has dimensions of 256 (time samples) × 256 (X direction samples) × 256 (Y direction samples). The dataset contains 50% missing traces, where 25% are located on the uniform grid and the remaining 25% are located on the non-uniform grid. Non-uniform samples are generated with a uniformly distributed distortion with the maximum offset equal to the spatial grid interval. Figure 3 shows the sampling geometry of the synthetic dataset. Missing samples, uniform samples, and non-uniform samples are represented by red circles, blue diamonds, and red triangles, respectively. The arrows indicate the original positions of the non-uniform samples. Figure 4 (a) shows the complete regular simulated data, and figure 4 (b) shows the data after 50% non-uniform missing traces with 25% are located on the uniform grid and the remaining 25% are located on the non-uniform grid. Figure 5 (a) is the reconstruction result by using conventional reconstruction method and figure 5 (b) is the corresponding error of figure 5 (a). because if the existence of deviation points, the S/N ratio is only 3.12dB, and some aliasing is introduced. Figure 6 (a) is the reconstruction result by using proposed reconstruction and regularization method and figure 6 (b) is the corresponding error of figure 6 (a). because of considering the influence of position of deviation points to the pre-defined reconstruction points, the S/N of final reconstruction is 12.1dB and greatly improved comparing the result by using conventional reconstruction method. The event becomes more continuous. This experiment demonstrates the effectiveness and robustness of the proposed approach for reconstructing non-uniformly sampled data.

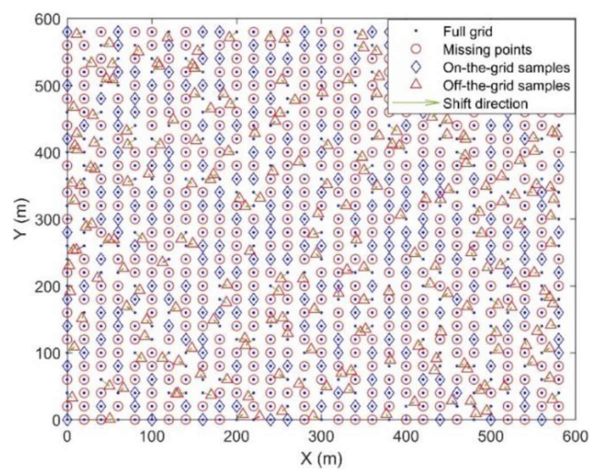

Figure 3—The sampling geometry of the simulated dataset

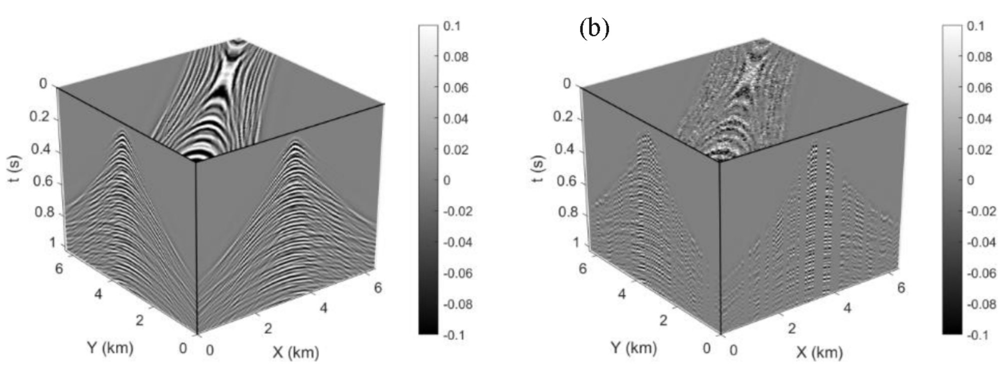

Figure 4—(a) the complete regular simulated data and (b) the data after 50% non uniform traces missing

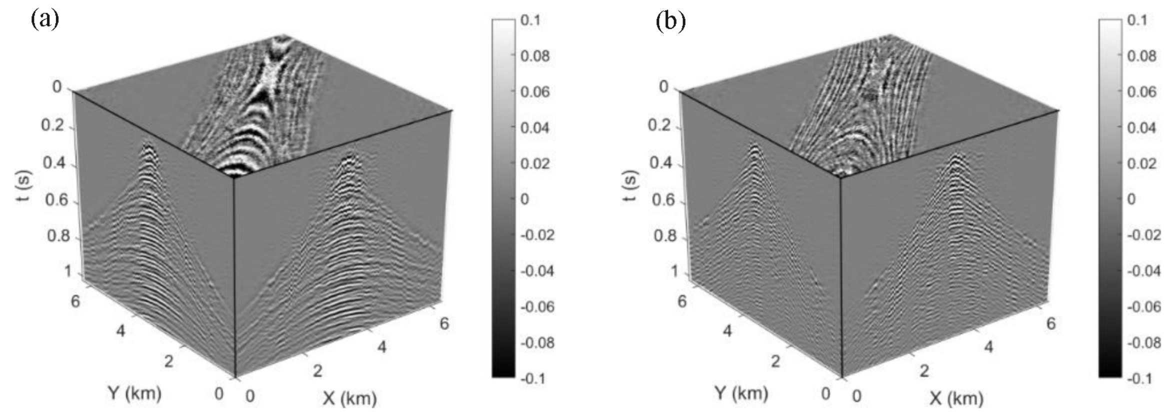

Figure 5—(a) the reconstruction result (S/N is 3.12dB) by using conventional reconstruction method and (b) the corresponding error

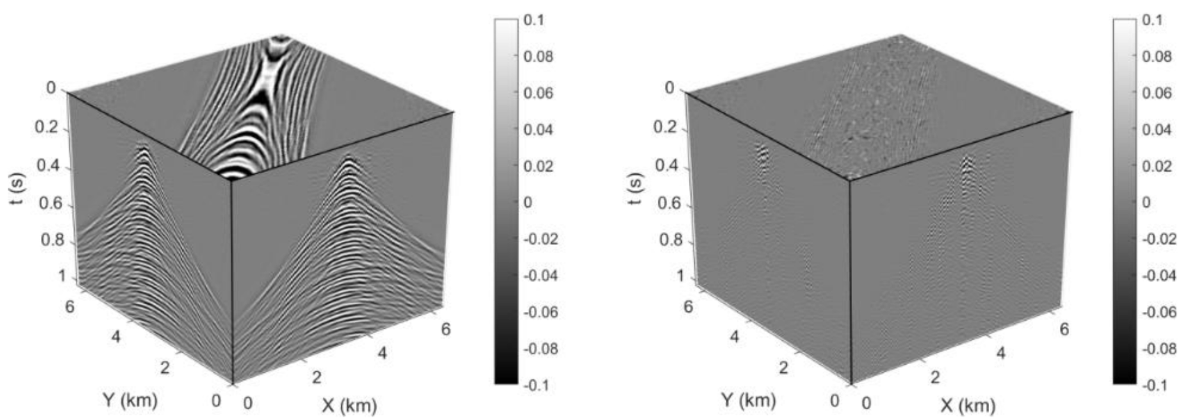

Figure 6—(a) the reconstruction result (S/N is 12.1dB) by using proposed reconstruction and regularization method and (b) the corresponding error

Next, we use a CS acquired field data example to demonstrate the performance of the proposed method. In the CS acquisition example, the source points and receiver points are both randomly decimated by 25% on the basis of a regular acquisition with grid size of 50m*50m, so the average source point interval and receiver point interval become about 66m after decimation, and the entire seismic data missing rate is about 44%. In order to recover the missing seismic data in both receiver line and source line directions, we choose to reconstruct the CS seismic data in cross spread. Figure 7a shows the source and receiver point distribution of one cross spread data before reconstruction. The blue points and red points denote the receiver points and source points respectively. We can see the receivers point and source points are irregularly distributed and some seismic data is missing. Figure 8a is the original CS seismic data. The reconstructed result is shown in figure 8b. From the comparisons, the proposed method can recover the missing data and make the seismic event more continuous. The reconstructed energy is also consistent with the original data. The source and receiver point distribution after reconstruction is shown in figure 7b. we can see the decimated source points and receiver points are recovered and the geometry becomes regular. Figure 9 shows the comparison of successive spread data in one cross spread before and after reconstruction. The proposed method can recover the missing data. Even for the worse situation of the entire data missing in one spread, the method still recovers the data very well. Also, we apply the method to a massive CS seismic data processing. In the test, 64 nodes are used. It takes 8 hours and 42 minutes to reconstruct 567GB of the original CS seismic data. The data size after reconstruction is 1.04 TB. From the analysis of computational speed, the proposed method has the industrial application capability to reconstruct the large-scale CS acquired seismic data.

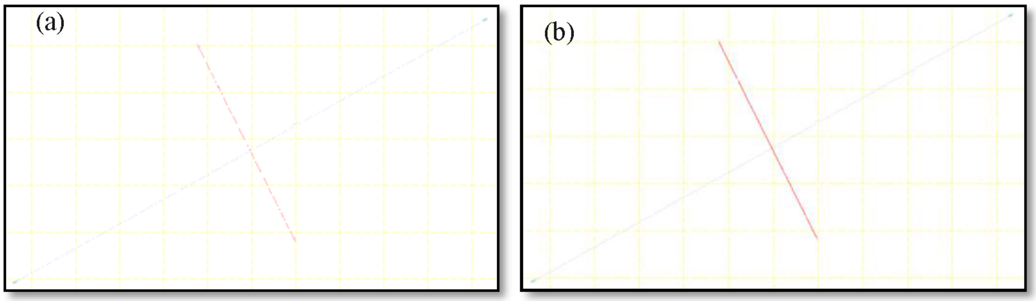

Figure 7—The sampling point distribution comparisons before (a) and after (b) reconstruction

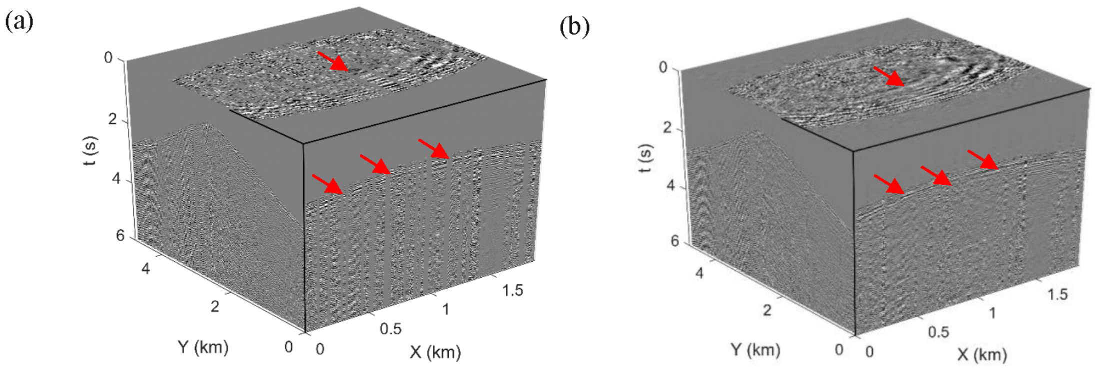

Figure 8—The comparisons of CS data before (a) and after (b) reconstruction

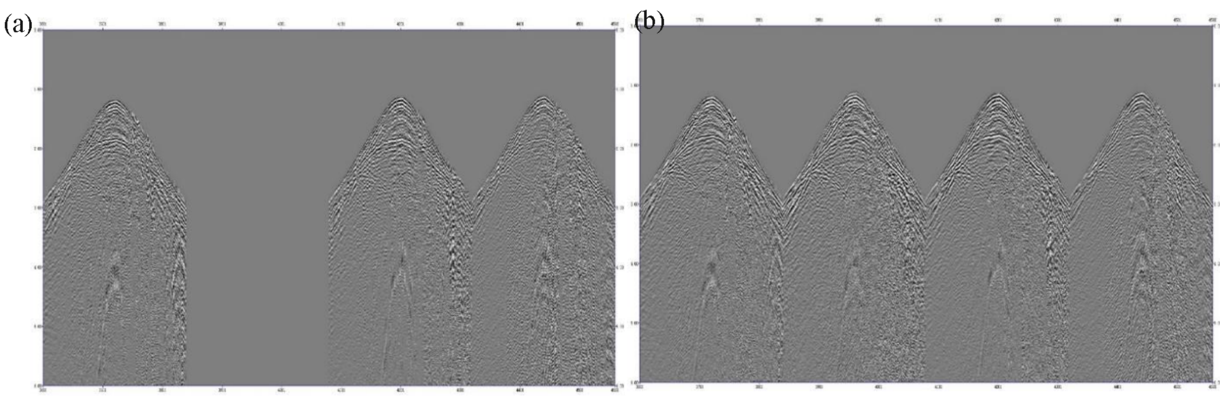

Figure 9—The comparisons of successive spread data in one cross spread before (a) and after (b) reconstruction

Conclusions

To tackle the low computational efficiency challenge of ISTA-based reconstruction method, a large-scale compressive sensing acquired data reconstruction method based on fast sparse inversion is proposed. In optimization algorithm aspect, we adopt the AISTA to improve the convergence speed and reconstruction accuracy. By considering the true positions of seismic samples, the non-uniform interpolation operator is designed to handle the deviation sampling points to optimally match the pre-defined reconstruction grids which makes the reconstruction results with less distortion and higher accuracy than the traditional methods based on approximate binning. Moreover, a parallel patching strategy is applied to 3D curvelet transform by segmenting a large volume of data into small patches to improve the computational efficiency. Through the application to massive CS acquired seismic data reconstruction, the proposed method can robustly recover the missing data and greatly improve the efficiency, which is proved to have the industrial application capability to reconstruct the large-scale CS acquired seismic data.

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